COMP396 Weekly Meeting Minutes

(Week 5)

Date & Time: Thursday, October 23rd, 2025, 13:30-15:00

Location: Harold Cohen Library/ Ground Floor Social Space

Note Taker: Zhang Olivia

Attendees: Zhang Olivia, Xue Yudan, Zhao Saria, Wong Haw

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1. Agenda:

- Review the progress of last week's tasks (D1–D3), focusing on the initial implementation of the strategy code and its chart application.
- Summarize key outcomes and suggestions from the meeting with the Supervisor.
- Based on the Supervisor's feedback and group discussion, finalize the strategy plan for Project Part 1 and Part 2.
- Plan the main tasks for the next stage (Week 6) and assign specific responsibilities.

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2. Discussion & Decisions:

2.1. Task Review and Result Confirmation

Team members reviewed the progress of last week's tasks (Week 4):

- **Strategy Implementation (D1):** The task was carried out as planned. Zhao Saria took the lead and completed the coding for Strategy 1 (S1). Strategies 2 (S2) and 3 (S3) have been added to the new plan for this week.
- **Visualization & Analysis Application (D2):** The task was completed. Through AI tools and hands-on practice, our team members gained a deeper understanding of the charts from

Lecture 6 and successfully applied them to visualize asset selection and strategy results (e.g., the PnL chart for S1).

- **Advanced Concepts & Report Preparation (D3):** The task was completed. Our team reviewed the relevant theoretical content of overfitting and developed a clearer understanding of the structure for the Design Document.

2.2. Supervisor Meeting Summary

Our team had a meeting with the Supervisor and achieved meaningful progress this week. The Supervisor provided valuable suggestions on several key issues, including:

- An evaluation of the current strategy selection and its overall rationality.
- Guidance on whether it is allowed to modify the BT396 framework and how to make adjustment to its underlying code.
- Suggested reading: *Trading Systems: A New Approach to System Optimization and Portfolio Construction* by Emilio Tomasini and Urban Jaekle.

2.3. Finalized Strategy Plan (Part 1 & Part 2)

After the meeting with the Supervisor, the group held an in-depth discussion and finalized the current strategy plans for the first two parts of the project:

Part 1： Single-Strategy Design

- **Asset Selection:** Based on the price patterns of 10 simulated datasets, we selected three assets with distinct characteristics for individual strategy development.
- **Strategy 1 (Trend-Following Type):** Targeting *Series 04* (clear upward trend), this strategy adopts a **Dual Moving Average Crossover approach.**
- **Strategy 2 (Passive Type):** Targeting *Series 05* (strong upward movement), this strategy applies a **Buy and Hold approach.**
- **Strategy 3 (Mean-Reversion Type):** Targeting *Series 09* (range-bound movement), this strategy applies a **Statistical Mean Reversion** approach, which is constructed based on the **Z-score**.

Part 2： Portfolio Strategy Design

- **Core Technique:** Our team decided to adopt **CVaR (Conditional Value at Risk)** as the optimization objective for portfolio construction.
- **Rationale:** Our team agreed that CVaR is superior to the traditional Mean-Variance (Markowitz) model, as it focuses more on minimizing tail risk, i.e., extreme loss scenarios.
- **Execution Plan:** The portfolio weights among the three strategies will be rebalanced every 20 days.

- **Technical Implementation:** The **SLSQP algorithm** will be applied to solve the constrained optimization problem.

2.4. Implementation Progress Update

- Zhao Saria has completed the coding and initial backtesting of Strategy 1 (S1).
- Result: S1 performed well in testing, achieving a PD Ratio of 5.093.

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3. Task Allocation:

To ensure the effective execution of the finalized strategy plans, our team has assigned the following tasks for the upcoming week:

ID	Module	Task Description	Owner	Due Date
E1	Strategy Implementation	Complete the coding and initial testing of Strategy 2 (Buy and Hold) and Strategy 3 (Z–score Mean Reversion).	Zhao Saria	Oct.30
E2	Code Review & Literature	1. Deeply read and analyze the reference book *Trading Systems: A New Approach to System Optimization and Portfolio Construction.* 2. Conduct a second review (Peer Review) of Zhao Saria's completed Strategy 1 (S1) code and related charts.	Wong Haw	Oct.30
E3	Portfolio Research & Support	1. Read literature such as *Optimization of Conditional Value-at-Risk* to gain a deeper understanding of CVaR–based portfolio optimization strategies. 2. Refine the detailed plan for Part 2 of the project. 3. Assist Zhao Saria in coding for Strategies 2 and 3.	Xue Yudan & Zhang Olivia	Oct.30

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4. Current Challenges and Issues

- **CVaR Optimization Implementation:** The main challenge has now shifted to Part 2. The next technical focus will be on how to effectively implement CVaR constraint optimization using the SLSQP algorithm with both accuracy and efficiency.

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5. Supervisor Question

- We will continue to follow the directions discussed with the Supervisor and focus on advancing the coding implementation of Strategies S1–S3, as well as the development of the Part 2 CVaR portfolio model.
- No urgent issues were reported. Our team aims to achieve significant progress before the next meeting.

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4. Next Meeting Arrangement

- **Date & Time:** Thursday, October 30th, 2025, 15:00-17:00
- **Location:** Sydney Jones Library / Abercromby Study Room 1
- **Key Agenda Items:**
  - Code review for Strategies S2 and S3 (E1).
  - Report on S1 code quality check (E2).
  - Update on CVaR research and Part 2 implementation details (E3).
  - Discussion on integration and parameter optimization for S1–S3 strategies.